where $\mathbf{R}$ is a column vector of two i.i.d. random normal numbers distributed as $\mathcal{N}(0,1)$, and

$$\boldsymbol{B}_t\boldsymbol{B}_t^T = \frac{2\gamma}{\beta}\int_0^t \exp(\boldsymbol{F}(t-s))\begin{bmatrix}0&0\\0&1\end{bmatrix}\exp(\boldsymbol{F}^T(t-s))\mathrm{d}s.$$

The solution to the slow part of our splitting is just

$$\begin{bmatrix}q(t)\\p(t)\end{bmatrix} = \begin{bmatrix}1&0\\-t&1\end{bmatrix}\begin{bmatrix}q(0)\\p(0)\end{bmatrix}.$$

Thus for a method taking one step of the fast system followed by a step of the slow system, using fixed step size $h > 0$, we can write the update scheme for $\boldsymbol{z} = [q,p]^T$ as

$$\boldsymbol{z}_{n+1} = \boldsymbol{X}\boldsymbol{z}_n + \boldsymbol{Y}\mathbf{R}_n,$$

for

$$\boldsymbol{X} = \begin{bmatrix}1&0\\-h&1\end{bmatrix}\exp(h\boldsymbol{F}), \qquad \boldsymbol{Y} = \begin{bmatrix}1&0\\-h&1\end{bmatrix}\boldsymbol{B}_h.$$

The stability of the method therefore depends solely on the spectral radius of the matrix $\boldsymbol{X}$. Denoting $\eta = \sqrt{\gamma^2 - 4\Omega^2}$, then we compute

$$\det(\boldsymbol{X}) = e^{-\gamma h}, \qquad \mathrm{tr}(\boldsymbol{X}) = \frac{\eta - h}{\eta}e^{(\eta-\gamma)h/2} + \frac{\eta+h}{\eta}e^{(-\eta-\gamma)h/2},$$

giving a stability condition

$$\left|\frac{\eta - h}{\eta}e^{(\eta-\gamma)h/2} + \frac{\eta+h}{\eta}e^{(-\eta-\gamma)h/2}\right| \leq 1 + e^{-\gamma h}.$$

Multiplying by $e^{\gamma h/2}$ gives

$$\left|\cosh\left(\frac{\eta h}{2}\right) + \frac{h}{\eta}\sinh\left(-\frac{\eta h}{2}\right)\right| \leq \cosh\left(\frac{\gamma h}{2}\right),$$

which reduces to the deterministic stability condition as expected in the limit $\gamma = 0$. For $\gamma > 0$, we can simplify to

$$\left|\cosh\left(\frac{\eta h}{2} - \mathrm{atanh}\left(\frac{h}{\eta}\right)\right)\sqrt{1 - \frac{h^2}{\eta^2}}\right| \leq \cosh\left(\frac{\gamma h}{2}\right).$$